

LVC v3: Dynamical Thawing Field with Geometry Coupling

Working Paper — Session Notes & Numerical Results
2025

Abstract

We present LVC v3, a refined formulation of the Luminal Velocity Correction framework in which the Hubble modification $T(z) = H(z)/H_{\text{LCDM}}(z)$ is derived from the dynamics of a thawing scalar field θ with a non-minimal geometry coupling $F(\theta)$. The key structural advance over LVC v2 is the replacement of a phenomenologically assumed $T(z)$ with a dynamically generated one: the field rolls from a frozen state near $\theta = \pi$ at $z \sim z_s$, passes through $\theta = \pi$ (the dip trigger), and refreezes by $z = 0$, producing $T(z=0) = 1$ automatically. We prove that $T < 1$ requires a geometry channel ($F(\theta) < 1$) rather than an energy channel, establish the unlock condition $\Lambda^4 \sim 2H(z_s)^2$, derive the $\Delta(z_s)$ constraint that fixes dip location at $z=1.1$, and quantify the SN χ^2 improvement vs BAO tradeoff. The No-Go theorem from LVC v2 is shown to persist in this setting: simultaneous SN improvement, D_M conservation at all BAO redshifts, and $\eta = 0.5$ remain mathematically incompatible within a single $T(z)$ function.

1. Motivation and Starting Point

LVC v2 achieved χ^2 improvements of $\Delta_{\text{SN}} = -167$, $\Delta_{\text{BAO}} = -8$ with $r_s = 148.38$ Mpc, CMB peak $l_1 = 220$ -221 matched, and $\eta = 0.670$ as a testable prediction distinct from LCDM ($\eta = 0.575$). The critical weakness was that $T(z)$ was constructed as a parameterized function (dip + boost with 6 free parameters) rather than derived from underlying dynamics. A reviewer could dismiss the entire framework as a sophisticated fitting exercise.

The central goal of LVC v3 is to close this gap: derive $T(z)$ from a scalar field ODE with a physically motivated potential and coupling structure, so that the dip and boost features emerge as predictions rather than inputs.

2. Theoretical Structure

2.1 The $T \geq 1$ No-Go and the Geometry Channel

The first structural result of this session is a proof that energy-based T definitions cannot produce $T < 1$. If one defines:

$$T(z)^2 = 1 + \rho_{\theta} / \rho_{\text{LCDM}}$$

then $T \geq 1$ is enforced by $\rho_{\theta} = (1/2)\dot{\theta}^2 + V(\theta) \geq 0$. This is a consequence of the standard Friedmann equation: adding any positive energy density always increases H . The SN improvement in LVC v2 requires $T < 1$ ($H < H_{\text{LCDM}}$) in the dip region, which is impossible in the energy channel.

The resolution is a geometry coupling. We define:

$$H(z)^2 = H_{\text{LCDM}}(z)^2 * F(\theta(z))$$

where $F(\theta)$ is a function of the scalar field value. When $F < 1$, $H < H_{\text{LCDM}}$ without violating energy conditions, because F encodes a modification of the effective gravitational coupling rather than an energy injection. The physically natural implementation is a non-minimal coupling action:

$$S = \int d^4x \sqrt{-g} \left[\frac{1}{2} F(\theta) R - \frac{1}{2} (d\theta)^2 - V(\theta) \right]$$

In the Einstein frame this produces $F(\theta)$ as the effective coupling function. The ansatz used for numerical exploration is:

$$F(\theta) = 1 - A * \exp(-(\theta - \pi)^2 / \sigma^2)$$

which produces $F < 1$ when θ passes through π . The parameters A (depth) and σ (width) are the two remaining physical degrees of freedom after the dip location constraint is applied.

2.2 The Scalar Field ODE

The scalar field equation of motion in FRW background with potential $V(\theta) = \Lambda^4 (1 - \cos \theta) + \delta \theta$ (Tilted Sine-Gordon) gives:

$$\ddot{\theta} + 3H \dot{\theta} + V'(\theta) = 0$$

$$V'(\theta) = \Lambda^4 (\sin \theta + 1) \text{ [with } \delta/\Lambda^4 = 1]$$

Converting to redshift z with $u = d\theta/dz$ and $d/dt = -(1+z)H d/dz$:

$$(1+z)^2 H^2 u' = -V'(\theta) + (1+z) H u [2H - (1+z) dH/dz]$$

where $H = H_{\text{LCDM}} * \sqrt{F(\theta(z))}$ is the self-consistent Hubble rate. This is a closed ODE system once initial conditions are specified.

2.3 The Attractor Analysis and Inflection Point Discovery

Numerical integration revealed that $\theta^* = 3.5\pi$ is NOT a stable attractor. Computing $V''(\theta^*) = \Lambda^4 \cos(\theta^*) = \Lambda^4 \cos(3.5\pi) = 0$ shows that $\theta^* = 3.5\pi$ is an inflection point (zero curvature), not a potential minimum. The field does not stop there but continues rolling — a transient crossing, not an attractor.

This is a key structural correction to LVC v2: the dip arises from a transient $\theta = \pi$ crossing (V' slope minimum), not from field trapping at an attractor. This is physically equivalent to the slow-roll to fast-roll transition in inflation exit, and is arguably more natural since it requires no fine-tuned minimum.

3. Thawing Field Initial Conditions

3.1 The Overdamping Problem and Its Resolution

A critical numerical finding: starting the field at $\theta_{\text{hi}} = 4\pi$, $z_{\text{hi}} = 10$ with $\Lambda^4 = 0.022$ ($= \Omega_{\text{ext}}$) produces a completely frozen field. At $z = 10$, $(1+z)^2 H^2 \sim 48400$, so the kinetic energy needed to produce $u_{\text{hi}} \sim 1.414$ requires $\rho_{\theta} \sim 48384 \gg \rho_{c,0} = 1$. The field is overdamped by Hubble friction at high redshift.

The resolution is the thawing scenario: the field is frozen ($u = 0$) at high z and begins rolling when $H^2 \sim \Lambda^4$. Setting $\Lambda^4 \sim 2 H(z_s)^2$ defines the unlock redshift z_s . For dip at $z = 1.1$, the required starting point is $z_s \sim 2-3$.

3.2 The Delta(z_s) Constraint

The initial offset $\Delta = \theta_s - \pi$ (how far the field starts from the dip trigger) determines the dip location. Imposing $z_{\text{dip}} = 1.1$ as a constraint yields Δ as a function of z_s . Numerical results:

z_s	$L4 = 2H(z_s)^2$	Δ^* (rad/pi)	T_{dip}	$T(z=0)$
1.8	14.57	0.1002	0.922	1.000
2.0	17.60	0.1527	0.922	1.000
2.2	21.06	0.2095	0.922	1.000
2.5	27.12	0.2987	0.922	1.000
2.8	34.32	0.3889	0.922	1.000
3.0	39.80	0.4484	0.922	1.000
3.5	56.07	0.5926	0.922	1.000

Table 1. $\Delta^*(z_s)$ constraint: initial field offset required to place dip at $z=1.1$, with $\Lambda^4 = 2H(z_s)^2$ and $u_s = 0$ (frozen initial condition). $T(z=0) = 1.000$ is achieved automatically for all z_s .

The automatic $T(z=0) = 1$ recovery is a key theoretical feature: the frozen initial condition $u_s = 0$ means the field starts with no kinetic energy, rolls through the dip, and refreezes as H grows large again at low z . No boundary condition tuning is required.

Δ is not a free parameter once z_s is fixed. The actual degree of freedom count reduces to (A, σ, z_s) — effectively 3 parameters, or 2 if z_s is determined by the unlock condition from external physics.

4. Supernova χ^2 Results

4.1 Methodology

We compute χ^2 against 36 binned Pantheon+ distance moduli (Brout et al. 2022), with analytic marginalization over absolute magnitude M :

$$\chi^2_{\text{marg}} = \sum_i w_i (\mu_{\text{th}}(z_i) + M^* - \mu_{\text{obs},i})^2$$

$$M^* = [\sum_i w_i (\mu_{\text{obs},i} - \mu_{\text{th}}(z_i))] / [\sum_i w_i], \quad w_i = 1/\sigma_i^2$$

ΛCDM baseline: $\chi^2_{\text{LCDM}} = 62.54$ (36 dof). $\Delta \chi^2 = \chi^2_{\text{LVC}} - \chi^2_{\text{LCDM}}$.

4.2 Dip-Only Results

With $F(\theta) = 1 - A \exp(-(\theta - \pi)^2 / \sigma^2)$ and the $\Delta(z_s)$ constraint ($z_{\text{dip}} = 1.1$ enforced), grid scan over (A, σ) :

A	σ	$\Delta \chi^2$	T_{dip}	$T(z=0)$
0.10	0.70	-9.5	0.949	1.000
0.20	0.70	-18.1	0.894	1.000
0.30	0.70	-25.8	0.837	1.000
0.40	1.50	-40.5	0.775	1.000

0.50	1.50	-46.1	0.707	1.000
0.60	1.50	-49.4	0.632	1.000

Table 2. SN Delta χ^2 for dip-only geometry coupling (36 Pantheon+ bins, M marginalized). $z_s = 2.5$ for all runs. $T(z=0) = 1.000$ in all cases.

4.3 BAO Constraint Violation

The SN improvement comes at a cost. For broad sigma (sigma = 1.5), $T < 1$ persists over $z \sim 0.5$ -2.5, causing D_M to deviate from LCDM over this range:

A	sigma	$D_M(0.35)/\text{LCDM}$	$D_M(0.57)/\text{LCDM}$	$D_M(1.0)/\text{LCDM}$	$D_M(2.34)/\text{LCDM}$
0.2	1.5	1.004	1.025	1.100	1.077
0.4	1.5	1.003	1.025	1.100	1.173

Table 3. D_M deviation from LCDM at BAO redshifts. BAO measurements require <1 -2% precision. Values above 1.01 indicate constraint violation.

5. No-Go Structure in LVC v3

The SN vs BAO tradeoff observed numerically has a clean mathematical explanation. The comoving distance integral

$$D_M(z) = \int_0^z dz' / H(z') = \int_0^z dz' / [H_{\text{LCDM}}(z') T(z')]$$

means that any $T < 1$ region between 0 and z increases $D_M(z)$ for all z beyond that region. SN improvement requires $T < 1$ near $z \sim 0.5$ -1.5. This necessarily inflates D_M at all higher redshifts including $z_{\text{BAO}} = 2.34$, unless a compensating $T > 1$ bump restores the integral. However, such a bump itself violates D_M conservation at intermediate BAO redshifts ($z \sim 0.57, 1.0$).

No-Go Theorem (LVC v3): For any single function $T(z)$, the conditions

- (i) $\Delta \chi^2_{\text{SN}} \ll 0$ (significant SN improvement)
- (ii) $D_M(z_{\text{BAO}}) = D_M^{\text{LCDM}}(z_{\text{BAO}})$ for all BAO redshifts
- (iii) $\eta = 0.5$

cannot be simultaneously satisfied. The obstruction arises because all three conditions are coupled through the same integral $D_M(z)$. This is the same structural result as LVC v2 No-Go, now reproduced from first principles in the geometry coupling framework.

6. The eta = 0.670 Prediction

The No-Go theorem means that within LVC, $\eta = 0.5$ is unachievable while maintaining SN and BAO simultaneously. Instead, LVC v2 predicts $\eta = 0.670$ as the value that naturally emerges when SN and BAO are jointly optimized. This is a testable prediction: LCDM predicts $\eta = 0.575$, LVC predicts $\eta = 0.670$. The difference $\Delta \eta = 0.095$ is in principle observable with next-generation surveys (DESI, Euclid, Roman).

7. Open Problems and Next Steps

Priority	Task	Status
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1 (urgent)	Phase B: Derive $F(\theta)$ from Brans-Dicke action	Not started
2	Full Pantheon+ fit (1701 SN, full covariance)	Not started
3	CAMB perturbation with $\sigma_T^{\text{eff}}(z)$ modification	Partial
4	Joint SN+BAO χ^2 Pareto frontier mapping	Started
5	Attractor scan: confirm $\Delta(z_s)$ stability	Partial
6	CMB high- l ($n=5-7$) Silk damping correction	Not started

The most critical next step is Phase B: deriving $F(\theta)$ from an action. The working ansatz $F(\theta) = 1 - A \exp(-(\theta-\pi)^2/\sigma^2)$ needs a Lagrangian origin. The natural candidate is the Brans-Dicke frame with non-minimal coupling $\xi \theta^2 R$, which produces $F(\theta) = 1 + \xi \theta^2 / M_{\text{Pl}}^2$. Matching this to the observed dip shape constrains ξ .

8. Summary

1. $T \geq 1$ is structurally enforced in the energy channel. $T < 1$ requires geometry coupling $F(\theta) < 1$.
2. The thawing scenario (frozen field unlocked at z_s) solves the overdamping problem and produces $T(z=0) = 1$ automatically.
3. The unlock condition $\Lambda^4 \sim 2H(z_s)^2$ and the $\Delta(z_s)$ constraint reduce the model to (A, σ, z_s) parameters.
4. SN improvement up to $\Delta \chi^2 \sim -50$ (36 bins) is achievable with dip at $z=1.1$.
5. BAO violation is unavoidable for significant SN improvement — the No-Go structure persists.
6. $\eta = 0.670$ remains the observable LVC prediction, distinguishable from ΛCDM $\eta = 0.575$.
7. $\theta^* = 3.5\pi$ is an inflection point, not an attractor. The dip is a transient crossing.